

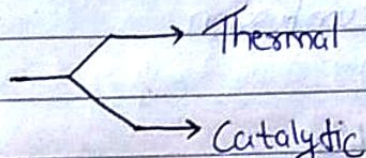
Fuels

- LCV (Lower calorific value)
- GCV (Gross calorific value)

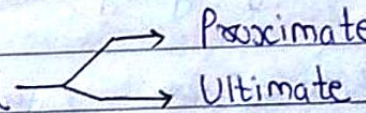
Determination

Bomb's calorimeter } Numericals
Boyl's calorimeter }

Octane no. & cetane no. (Applications)

⇒ Cracking 

Solid fuel → Coal

Numericals → Analysis of coal 

Carbonisation of fuel

- HTC (High temp carbonisation)
- LTC (Low temp carbonisation)

Manufacture of coke : Otto Hoffmann's oven

Combustion of fuel : Numericals

⇒ Date : 9 Jan '21. Topic : Fuels

Fuels : Substances that release ^{energy} when they undergo combustion.

Calorific : Amount of heat given to increase the temp of 1g of fuel by 1°C

Calorific value

HCV / GCV

The amt. of heat liberated when unit / volume, unit mass of fuel has been burnt completely & the products of combustion are cooled down to room temp.

LCV / NCV

Net heat produced when unit mass / volume of fuel is burnt completely and combustion products are escaped.

Calorific value : Amount of heat generated when unit mass / volume of fuel undergo combustion.

★ Relationship between GCV & LCV

a) When mass of Hydrogen is present

$$LCV = HCV - \text{Latent heat of vaporisation}$$

$$LCV = HCV - (9H \cdot 587)$$

9H = weight of water

H = weight of hydrogen

b) When hydrogen content is in %

$$LCV = HCV - (0.09 H \times 587)$$

★ Dulong's formula

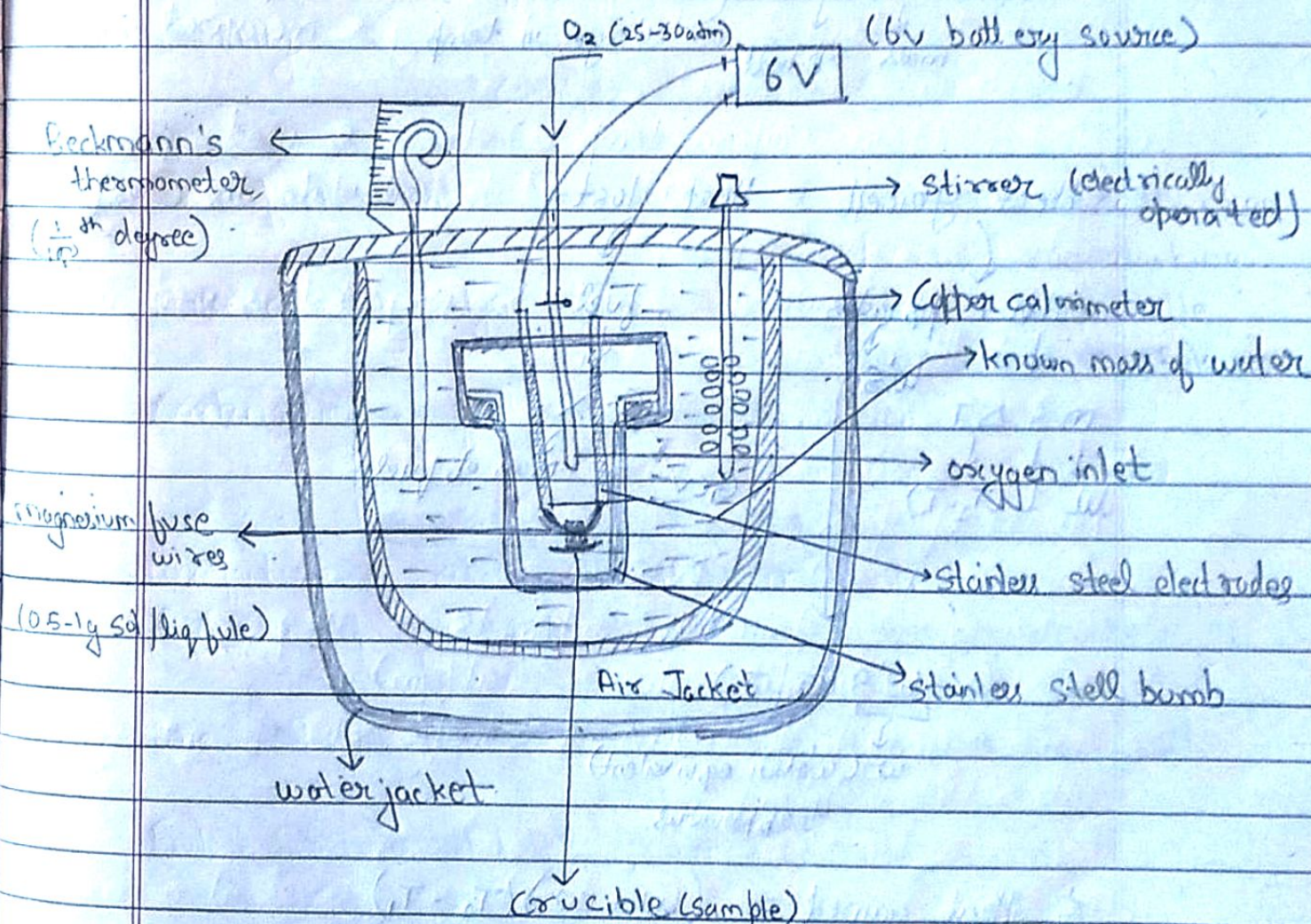
$$GCV = \frac{1}{100} [8080 \cdot \%C + 34500 (\%H - \frac{\%O}{8}) + 2240 \cdot \%S]$$

unit: cal/gm

$\%C = \% \text{ of carbon}$
 $\%H = \% \text{ of hydrogen}$
 $\%O = \% \text{ of oxygen}$

in $\left(\frac{\%H}{1} - \frac{\%O}{8} \right)$
 Hydrogen which didn't react with oxygen.
 Hydrogen which made water

⇒ Date: 12 Jan '21



Calculations

Gross cal. value of fuel = known. (C)

mass of sample = x gms

mass of water = W gms

Initial temp = T_1

Final temp = T_2

mass of apparatus = W' gms

$$\text{Heat} = M \cdot S \cdot \Delta T$$

$\downarrow$ mass $\downarrow$ sp. heat $\rightarrow$ change in temp.

Heat gained = Heat lost

water + apparatus

$$M \cdot S \cdot \Delta T$$

$\downarrow \quad \downarrow \quad \downarrow$
 $W \cdot S \cdot (T_2 - T_1)$

fuel

$$C \cdot X$$

$\swarrow \quad \searrow$
 GCV mass of fuel

$$W' \cdot S \cdot (T_2 - T_1)$$

w (water equivalent)
of apparatus.

$$\star \text{ Heat gained} = (W + w)(T_2 - T_1)$$

⇒ Corrections

The extra amount of heat produced due to spark in fuse wire needs to be subtracted

reaction (CA)

$$\begin{array}{l} \text{S} + \text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{SO}_3 \\ \text{S} + \text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \Delta H \end{array} \quad \left. \vphantom{\begin{array}{l} \text{S} + \text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{SO}_3 \\ \text{S} + \text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \Delta H \end{array}} \right\} \text{exothermic rxn}$$
$$\text{N} + \text{H} + \text{O}_2 \rightarrow \text{HNO}_3 + \Delta H$$

⇒ The extra amount of heat which is liberated, due to exothermic rxn of formation of acid, needs to be subtracted.

Initially the water is at room temp, but when it is transferred into calorimeter, it loses some temperature, as the surrounding components are at low temp than that of water. So the actual temp of water is T_c . ~~So~~th

So the diff in temp (ΔT) = $(T_2 - T_1) + T_c$

$$C = (W + w) (\Delta T + T_c) - (C_f + C_A)$$

Construction / Working

- It consists of a stainless steel container (bomb)
- The fuel is placed in a crucible in bomb.
- Fuse wire is used to ignite the fuel, connected to a 6V battery.
- The bomb is at a pressure of 25-30 atm with O_2 .
- As the key is pressed a spark is produced which ignites the fuel and fuel starts burning.
- Due to combustion of fuel heat is generated which is transferred to the outside container containing water.
- Due to this temp of water starts rising & measured with Beckmann's Thermometer.
- To maintain uniform temp a stirrer is used.
- Rise in temp is (ΔT) used to calculate the calorific value as given below.
- To maintain insulation a water & air jacket is used to surround the outside container.

⇒

Amount of fuel = x gamount of water = W gwater eq. of calorimeter components = w g C = High calorific value (cal/gm) T_1 = initial temp T_2 = Final temp→ Heat produced by 1 g fuel = C Total heat produced by x g fuel = $C \cdot x$

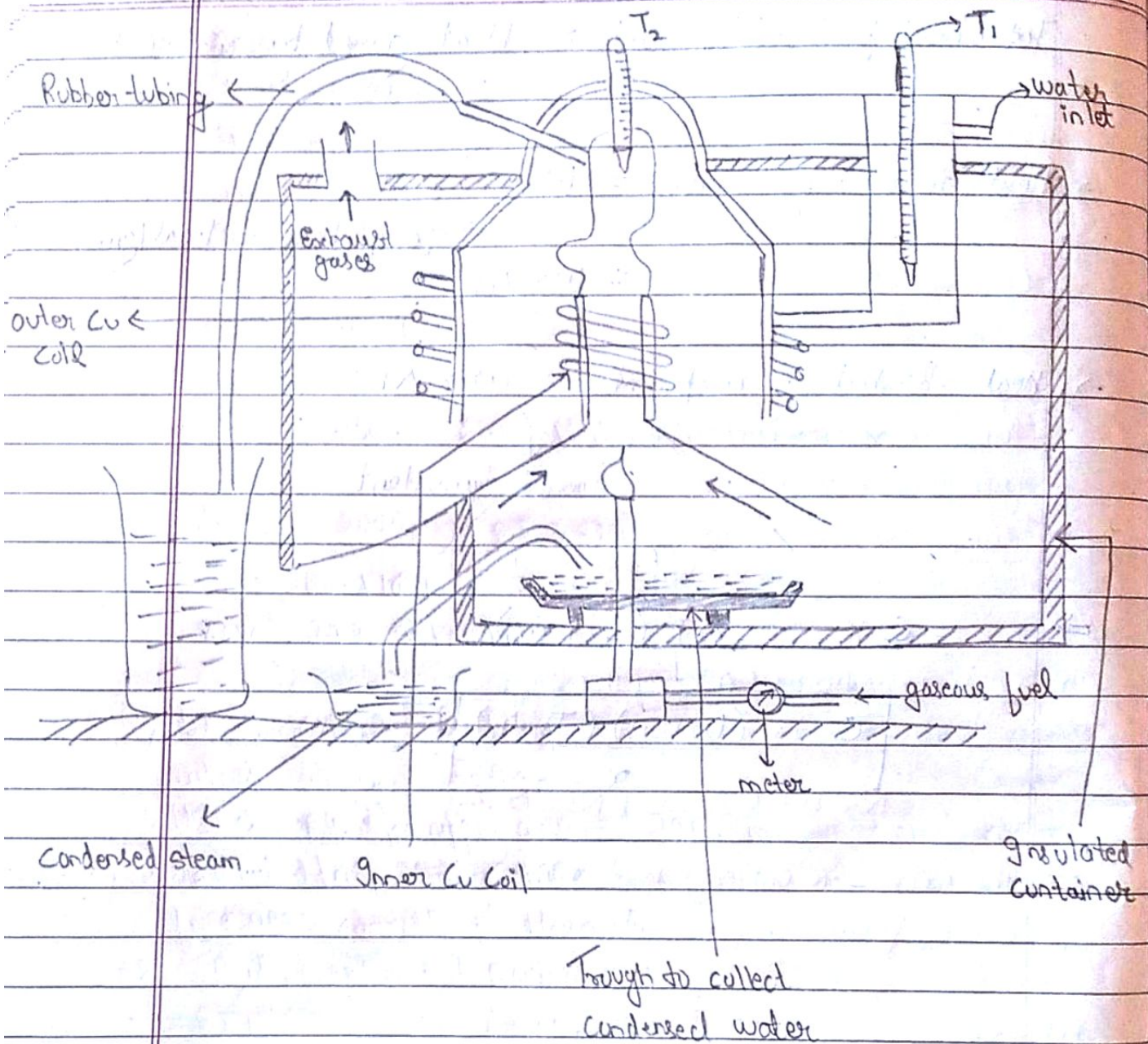
Total heat produced by fuel = Heat gained by water & calorimeter.

→ Heat absorbed by water = $W \times S \times \Delta T$
 $\hookrightarrow$ specific heat = 1 cal/gm
 $= W \times \Delta T$

→ Heat absorbed by components = $w \times \Delta T$
 $\hookrightarrow$ water equivalent

$\Rightarrow C \cdot X = \Delta T (W + w)$

$$C = \frac{(W + w) (T_2 - T_1)}{x}$$



Boy's Gas Calorimeter

⇒ It is used for determination of calorific value of gaseous and volatile liquid fuels.

Principle : The gaseous fuel is burnt in a vessel at a constant rate in such a manner that the entire heat is absorbed by the water which is circulated in a chimney around the burner. From the vol. of gas fuel burnt, the

volume of water collected and the rise of temp of water & the calorific value of fuel is determined.

$$\text{Heat lost} = \text{Heat gained}$$

Calculation

Vol of fuel present at STP in given time $t = V \text{ m}^3$

Wt of water passed through coil in given time $t = W \text{ kg}$

Initial temp = T_1

Final temp = T_2

Weight of water condensed = $W' \text{ kg}$

→ Let HCV of fuel = $(GCV) = C \text{ kcal/m}^3$

→ Sp heat of water = $1 \text{ kcal/kg}^\circ\text{C}$

$$\text{Heat gained} = \text{Heat lost}$$

↓
water

↓
fuel

$(ms\Delta T)$

$(ms\Delta T)$

$$= CV$$

$$W \times 1 \times (T_2 - T_1) = CV$$

$$C = \frac{W(T_2 - T_1)}{V}$$

$$LCV = GCV - \left(\frac{W'}{V} \right) \times S\Delta T$$

→ wt of water condensed per m^3 of fuel burnt

Q, $V = 0.1 \text{ m}^3$

$T_1 = 26^\circ \text{C}$

$T_2 = 36^\circ \text{C}$

$W = 26 \text{ kg}$

$W' = 0.03 \text{ kg}$

$$C = \frac{26}{0.1} (36 - 26)$$

$\therefore mV = 1 = 2600 - \text{Kcal}$

Cal (L) $LCV = \frac{2600 - 0.03 \times 587}{0.1}$
 $= 2600 - 176.1$

$LCV = 2423.9 \text{ Kcal/m}^3$

Q, wt of coal = 0.920 g

wt of water = 550 gm

Water eq of apparatus = 2200 gm

rise in temp = 24.2°C

fuse wire correct = 10 cal

Acid corr = 50 cal

Cal sample = $C = 93\%$

$H = 6\%$

ash = 1%

GCV, LCV

$(T_2 - T_1) \times 1 \times 10 = 7168.47$

$$C = \frac{(W + w) (\Delta T) - (C_f + C_a)}{x}$$

$C = \frac{(550 + 2200) (24.2) - (10 + 50)}{0.920}$

$= 7168.47 \text{ cal/gm}$

$$LCV = GCV - \left(\frac{9H}{100} \times 587 \right)$$

$$= 7162.47 - 0.09 \times 6 \times 587$$

$$= 6851.49 \text{ cal/gm}$$

Qy Fuel A: C = 80%, O = 5%, S = 25%,

H = 10.4%, Ash = 2.1%.

Fuel B: C = 75%, O = 12.1%, S = 3.2%, H = 5.2%.

Ash = 4.5%.

$$\text{Fuel A: } GCV = \frac{1}{100} \left[80 \times 80 \times 80 + 34500 \left(\frac{10.4 - 5}{8} \right) + 2240 \times 2.5 \right]$$

$$= \frac{1}{100} [646400 + 337237.5 + 5600]$$

$$= 9892.375 \text{ cal/gm}$$

$$LCV = 9892.375 - 0.09 \times 10.4 \times 587$$

$$= 9379.9234 \text{ cal/gm}$$

$$\text{Fuel B: } GCV = \frac{1}{100} \left[80 \times 75 + 34500 \left(\frac{5.2 - 12.1}{8} \right) + 2240 \times 3.2 \right]$$

$$= 2471.67 \text{ cal/gm}$$

$$LCV = 2471.67 - 0.09 (5.2) \times 587$$

$$= 2276.85 \text{ cal/gm}$$

★ Octane number & Cetane number

→ Ignition temp: Minimum temp at which combustion is self supporting. Also known as spontaneous ignition Temp. (SIT)

→ Compression ratio: Indicates extent of compression of fuel & air mixture, compression is done by piston.

$$CR = \frac{V_1}{V_2} > 1$$

V_1 → Gas vol. in the cylinder at the end of suction stroke.

V_2 → Vol. at the end of compression stroke of the piston.

→ Knocking: Sharp metallic sound produced in internal combustion engine due to immature ignition of air fuel mixture.

→ The C.R at which fuel tends to knock is called critical combustion ratio. (CCR)

→ The temperature at which knocking is taking place is called SIT (Spontaneous ignition temp)

→ Knocking results in:

① Decreased power output

② Mechanical Damage

→ Factors on which knocking depends:

① Engine design

② Running conditions

③ Chemical structure of fuel hydrocarbons

④ Straight chain hydrocarbons knock severely, so they have lowest O-N

VVIP

⇒ How knocking depends on chemical structure of fuel

→ As branching increases knocking decreases.

n -alkane > monosubstituted alkane > cycloalkanes > alkenes > polysubstituted alkenes > aromatic

→ For straight chain hydrocarbons, less the carbon atoms, less will be the knocking.

n -butane < n -pentane < n -hexane

⇒ Resistance offered by gasoline towards ^{knocking} is expressed on a scale called octane number.

⇒ Lesser the knocking, more will be octane number, hence better will be the fuel.

	A	B	C	D	E
O.N ⇒	70	60	50	80	20

	O.N	Knocking
fuel → n -heptane	0	100
→ iso-octane	100	0

→ Anti knocking agents

eg- ① TEL → Tetraethyl lead $(C_2H_5)_4 Pb$

② Diethyl telluride $(C_2H_5)_2 Te$

★ lead & lead dioxide decreases the engine life.
(Pb) (PbO_2)

⇒ Benzene is also used to decrease knocking. It is harmful as it is carcinogenic.

Cetane number (diesel engine)

Cetane no. is based on compression ignition.

Ignition delay

Time taken for vapourisation for fuel droplets

It is due to the time taken for vapourisation of fuel droplets and raising the temp of vapors to ignition temperature.

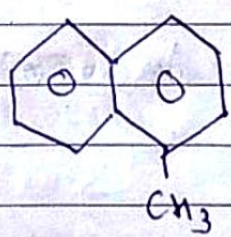
Ignition delay is the time interval b/w start of fuel injection & its ignition.

Lower the ignition delay, more will be the cetane number & better will be the fuel.

It is the measure of the ease, with which the fuel will ignite.

A	B	C	D	E	Best fuel
Cetane - 80	70	30	10	90	
α -methyl naphthalene 20					
80					

fuel $\rightarrow$ Cetane - $C_{16}H_{34}$ (n-hexadecane) - 100
 $\rightarrow$ α -methyl naphthalene



ignition delay

cetane 0

α -methyl
naphthalene 100

Q, Power petrol? (NW)

Q, How cetane number depends on the chemical structure of the diesel fuel.

⇒ $n\text{-alkane} > \text{cycloalkanes} > \text{alkenes} > \text{branched alkane} > \text{aromatic}$

max cetane number ← ignition delay increases → min cetane number

ignition quality decreases →

Cetane number decreases →

⇒

NW// Diagram & Power Petrol

⇒ Power petrol

Power petrol is a kind of premium petrol & normal is unleaded. The main difference between regular and power fuel in the power or premium fuels have a higher number of octane number. In midgrade petrol, the number of octanes varies from 88 to 90. In premium of power petrol, there is higher number of octane i.e. 91 to 94.

- Q) Calculate the gross & net calorific values of coal having
 $C=85\%$, $H=8\%$, $S=1\%$, $N=2\%$, ash = 4%.
 Latent of steam = 587 cal/gm

$$HCV = \frac{1}{100} [8080 C + 34500 [H - \frac{O}{8}] + 2240 S] \text{ kcal/kg}$$

$$HCV = \frac{1}{100} [8080 \times 85 + 34500 [8 - \frac{0}{8}] + 2240 (1)] \quad \text{--- (1)}$$

$$= \frac{1}{100} [686800 + 27600 + 2240]$$

To find comp. of %O :-

$$\%O = 100[85 + 8 + 1 + 2 + 4]$$

$$O = 0\%$$

$$GCV \quad HCV = \frac{1}{100} [686800 + 27600 + 2240]$$

$$HCV = 9650.40 \text{ kcal/kg}$$

$$NCV / LCV = HCV - (9H \times 587)$$

$$LCV = 9650.4 - (9 \times 8 \times 587)$$

$$LCV = 9227.76 \text{ kcal/kg}$$

- Q) A coal has the following composition by weight $C=90\%$, $O=30\%$,
 $S=0.5\%$, ash = 2.5%, given $NCV/LCV = 8490.5 \text{ kcal/kg}$
 calculate the % of H & HCV of fuel.

$$NCV = HCV - 0.09H \times 587$$

$$HCV = NCV + 0.09H (587)$$

$$= (8490.5 + 0.09 \times 587 H)$$

$$HCV = (8490.5 + 52.83H) \text{ --- (i)}$$

$$\text{also, } HCV = \frac{1}{100} [8080(90) + 34500(H - \frac{3}{8}) + 2240 \times 0.5]$$

$$= [7153.8 + 345H] \text{ --- (ii)}$$

from (i) & (ii) :-

$$7153.8 + 345H = 8490.5 + 52.83H$$

$$292.2H = 8490.5 - 7153.8 = 1336.7$$

$$\% H = \frac{1336.7}{292.2} = 4.574\%$$

Q) In an exp of Bomb calorimeter, a solid fuel of 0.90 g is burnt. It is observed that increase in temp is 3.8°C of 4000 g of water. The fuel contains 1% of H. Calculate the HCV & LCV of water eq. is 385 g & latent heat of steam is 587 cal/g.

$$m = 0.90\text{g}, \Delta T = 3.8^\circ\text{C}, W = 4000\text{g}, w = 385\text{g}, L = 587\text{cal/g}$$

$$\% H = 1$$

$$\Rightarrow HCV = \frac{(W + w)(\Delta T)}{m}$$

$$HCV = \frac{(4000 + 385)(3.8)}{0.90}$$

$$HCV = \frac{16663}{0.90} = 18514.5 \text{ cal/g}$$

$$HCV = 18514.5 \text{ cal/g}$$

$$LCV = HCV - [0.09H \times 587]$$

$$\text{LCV} = 18514.5 - [0.09(1) \times 587]$$

$$= 18514.5 - 52.83$$

$$\boxed{\text{LCV} = 18461.67 \text{ cal/g}}$$

Q) A sample of coal containing 89% C, 8% H, 3% ash when the coal sample was tested in laboratory in bomb calorimeter, the following data were obtained.

Weight of coal burnt = 0.85g

Weight of water taken = 650g

Water eq. of bomb calorimeter = 2500g

$\Delta T = 2.5^\circ\text{C}$

Cooling correction = 0.03°C

Fuse wire correction = 10 cal

Air correction = 50 cal

Latent heat of steam = 587 cal/g

Calculate GCV & NCV of coal

$$\text{GCV} = \frac{(W + W_w)(\Delta T + C_c) - (C_f + C_A)}{m}$$

$$= \frac{(650 + 2500)(2.5 + 0.03) - (10 + 50)}{0.85}$$

$$= \frac{(3150 \times 2.53) - (60)}{0.85} = \frac{7969.5 - 60}{0.85}$$

$$\text{GCV} = \frac{7909.5}{0.85} = 9305.29$$

$$\boxed{\text{GCV} = 9305.29 \text{ cal/g}}$$

$$\text{NCV} = \text{GCV} - [0.09 H \times 587]$$

$$= 9305.29 - [0.09(8) \times 587]$$

$$NCV = 9305.09 - 422.64$$

$$NCV = 8882.65 \text{ cal/g}$$

- Q) A sample of coal contains C = 83%, H = 6%, Ash = 1%.
 weight of coal burnt = 0.92 g
 weight of water taken = 550 g
 water eq. of calorimeter = 2200 g
 $\Delta T = 2.42^\circ C$

$$\text{Fuse wire correction} = 10 \text{ cal}$$

$$\text{Acid correction} = 50 \text{ cal}$$

calculate GCV & NCV of coal.

$$\text{Latent heat of steam} = 580 \text{ cal/g}$$

Acc. to ~~Boyle's~~ Bomb's calorimeter

$$GCV = \frac{(W + w)(\Delta T) - (C_f + CA)}{m}$$

$$= \frac{(550 + 2200)(2.42) - (10 + 50)}{0.92}$$

$$= \frac{2750 \times 2.42 - 60}{0.92} = \frac{6655 - 60}{0.92}$$

$$GCV = \frac{6595}{0.92} = 7168.5$$

$$GCV = 7168.5 \text{ cal/g}$$

$$NCV = GCV - [0.09H \times 580]$$

$$= 7168.5 - [0.09 \times 6 \times 580]$$

$$= 7168.5 - 313.2$$

$$NCV = 6855.3 \text{ cal/g}$$

Q) The following data is obtained in a bomb calorimeter experiment:

Weight of crucible = 3.649 g

weight of crucible + fuel = 4.678 g

weight of apparatus = 5216.33 g

mass of specific heat = 0.098

observed rise in temp = 2.3°C

cooling correction = 0.047°C

acid correction = 62.6°C

Fuse wire correction = 3.8°C

Cotton thread correction = 1.6 cal

Calculate GCV of fuel sample if the fuel contains 6.5% H, determine the NCV.

Water taken in calorimeter = 2200 g

$$\begin{aligned}\text{mass of fuel} &= 4.678 - 3.649 \\ &= 1.029\end{aligned}$$

$$\begin{aligned}\text{Water eq} &= \text{mass of apparatus} \times \text{mass of sp. heat} \\ &= 5216.33 \times 0.098 \\ &= 570 \text{ cal/g}\end{aligned}$$

$$\text{GCV} = \frac{(W + w)(\Delta T + C_c) - (C_f + C_a + C_T)}{m}$$

$$\text{GCV} = \frac{(2200 + 570)(2.3 + 0.047) - (3.8 + 62.6 + 1.6)}{1.029}$$

$$= \frac{2770 \times 2.347 - (68)}{1.029}$$

$$= \frac{6501.19 - 68}{1.029}$$

$$\text{GCV} = 6251.88 \text{ cal/g}$$

$$\begin{aligned}
 \text{NCV} &= \text{GCV} - [0.09 \text{ H} \times 587] \\
 &= 6251.88 - [0.09(6.5) \times 587] \\
 &= 625.88 - 343.395
 \end{aligned}$$

$$\boxed{\text{NCV} = 5908.485 \text{ cal/g}}$$

★ Cracking

Decomposition of larger hydrocarbon into the smaller chain hydrocarbon is called cracking.

Cracking can take place if heat is applied or by the use of catalyst.

⇒ Cracking is of two types $\begin{cases} \rightarrow \text{Thermal cracking} \\ \rightarrow \text{Catalytic cracking} \end{cases}$

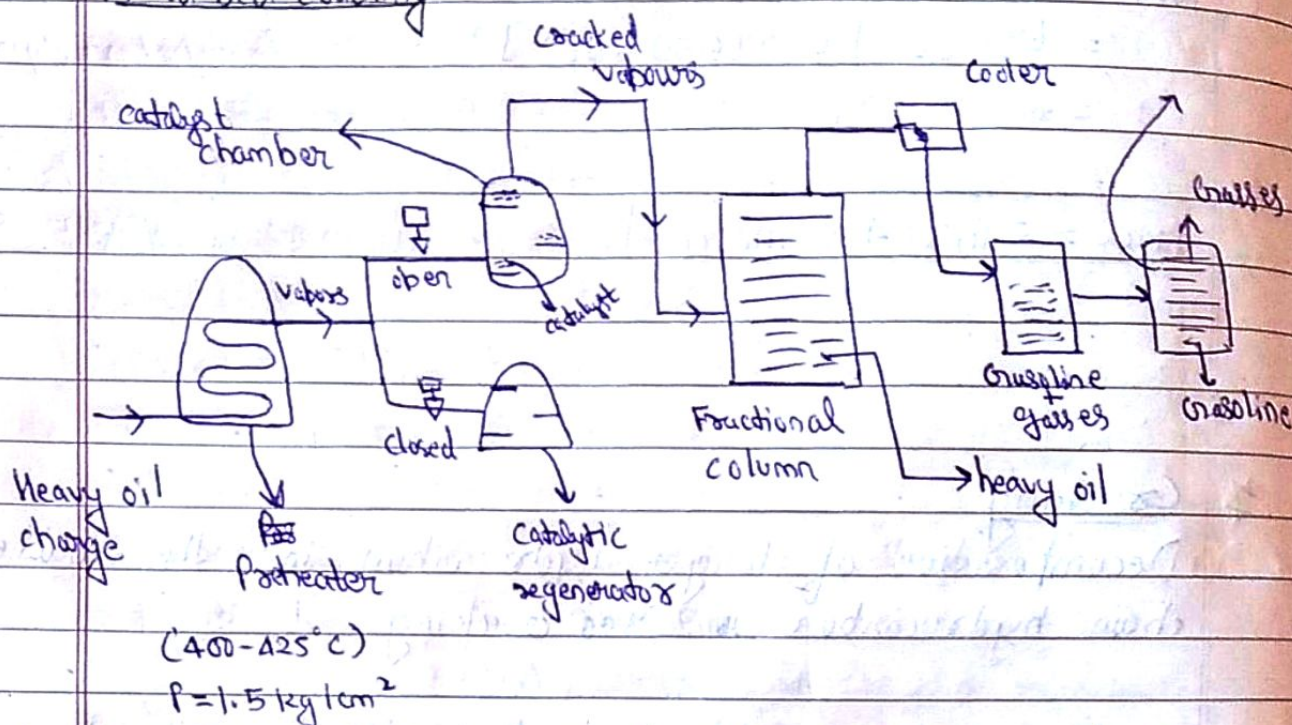
i) Thermal cracking

$\begin{cases} \rightarrow \text{Liquid phase} \\ \rightarrow \text{Vapour phase} \end{cases}$

ii) Catalytic cracking

$\begin{cases} \rightarrow \text{Fixed bed cracking} \\ \rightarrow \text{Moving bed cracking} \end{cases}$

Fixed bed cracking



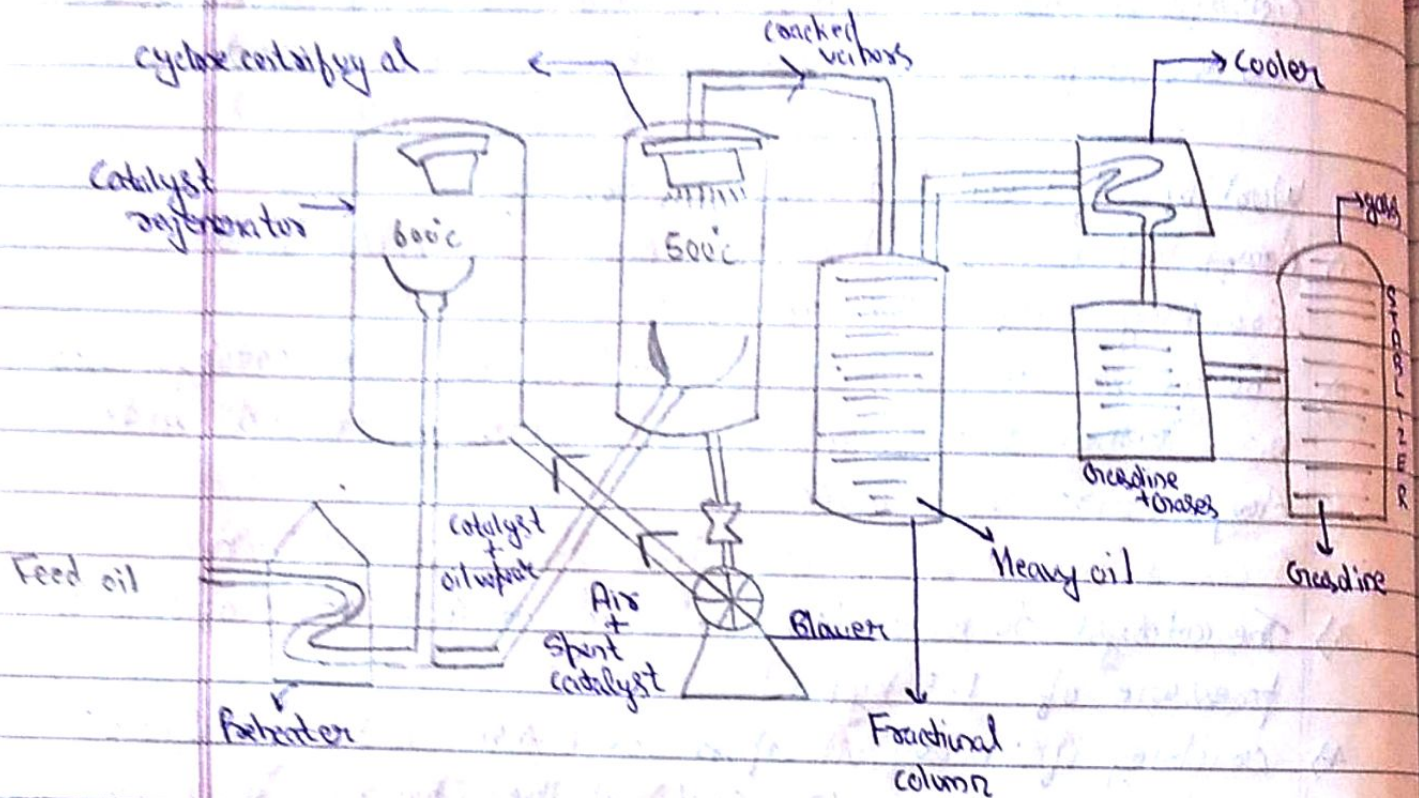
- ① Vapours of heavy oil are heated in presence of catalyst due to which better yield of petrol is obtained.
- ② Heavy oil is vapourised by heating in an electrical heater.
- ③ Generally catalysts used are bauxite, zeolite, crystalline etc.
- ④ The cracked gases are taken out from the top of the chamber and is allowed to pass into fractionating tower, where gasoline is collected.
- ⑤ During cracking free carbon is also formed which deposits on catalyst, then flow of vapours of heavy oil is passed over the second set of chamber.

⑥ Catalyst in earlier chamber is regenerated by burning the carbon deposits with the help of air & used.

⇒ Working of fixed Bed catalytic cracking

- 1) Heavy oil is passed through a heater and heated to cracking temp ($420-450^{\circ}\text{C}$)
- 2) Charge is then passes to catalytic chamber. The catalyst used is either silica-aluminium gel or bauxite mixed with clay & zirconium oxide.
- 3) The catalyst tower is also maintained at $420-450^{\circ}\text{C}$ and at a pressure of 1.5 kg/cm^2
- 4) cracking of heavy oil vapours takes place in the catalytic chamber & about (30-40%) of the charge is converted into low molecular weight hydrocarbons.
- 5) Carbon formed gets absorbed by catalyst thereby deactivating it.
- 6) The cracked vapours are passed through the fractional column, where gasoline vapours are recovered from the top & heavy oil gets condensed at the bottom.
- 7) The gasoline vapours are subjected to the cooler where gasoline along with some other gaseous products gets condensed.
- 8) Thereafter, the gasoline is sent to the stabilizer, where the dissolved gases are removed from the top & pure gasoline is obtained, having the desired boiling.

Moving bed catalytic cracking



⇒ In this type of cracking, the catalyst is in the form of fine powder, which flows down to the cracking chamber through a hopper. The catalyst get mixed with the heavy oil vapours and behave much like a fluid which can be circulated in gas streams. The plants for fluid bed or moving bed catalyst cracking consists of a reactor & regenerator placed side by side.

★ Analysis of fuel → Proximate analysis
→ Ultimate analysis

Proximate

- % of moisture
- % of volatile ~~metals~~ matter
- % of ash
- % of fixed C

C%, H%, N%, S%

⇒ Proximate analysis

10 gm $\xrightarrow[\Delta]{\text{-moisture}}$ 8 gm $\xrightarrow[\Delta]{\text{-volatile}}$ 7 gm $\longrightarrow$ 5 gm ash

$$\% \text{ of moisture} = \frac{\text{Loss in wt. due to moisture}}{\text{Wt. of coal}} \times 100 \quad \left(\frac{2}{10} \times 100 \right)$$

$$\% \text{ of volatile comp.} = \frac{\text{Loss in wt. due to volatile matter}}{\text{Wt. of coal}} \times 100 \quad \left(\frac{1}{10} \times 100 \right)$$

$$\% \text{ of ash} = \frac{\text{Wt. of residue}}{\text{Wt. of coal}} \times 100 \quad \left(\frac{5}{10} \times 100 \right)$$

$$\Rightarrow \boxed{\% \text{ of fixed C} = 100 - (\% \text{ of moisture} + \% \text{ of volatile matter} + \% \text{ of ash})}$$

	Initial wt.	wt. after removal of moisture	after vol. matter	residue	
A	2.5	2.41	1.528	0.245	(51.32)
B	5.0	4.8	3.38	0.075	(66.1)

$$\begin{aligned} \% \text{ fixed C (A)} &= 100 - \left(\frac{0.09}{2.5} \times 100 + \frac{0.832}{2.5} \times 100 + \frac{0.245}{2.5} \times 100 \right) \\ &= 100 - 48.68 \\ &= 51.32 \% \end{aligned}$$

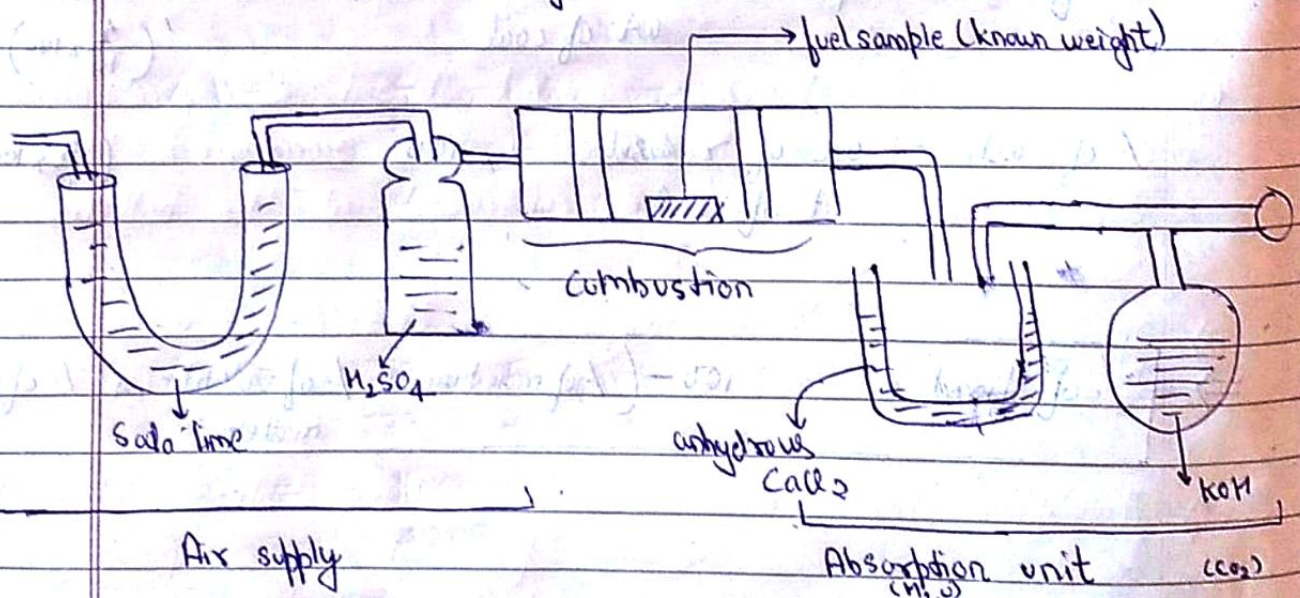
$$\begin{aligned} \% \text{ fixed C (B)} &= 100 - \left(\frac{0.2}{5} \times 100 + \frac{1.42}{5} \times 100 + \frac{0.075}{5} \times 100 \right) \\ &= 100 - 33.9 \\ &= 66.1 \% \end{aligned}$$

(Fuel B is better)

★ Ultimate analysis

% C, % H, % N, % S

Kjeldahl's



	Initial wt.	wt. after removal of moisture	after vol. matter	residue	
A	2.5	2.41	1.528	0.245	(51.32)
B	5.0	4.8	3.38	0.075	(66.1)

$$\begin{aligned} \% \text{ fixed C (A)} &= 100 - \left(\frac{0.09}{2.5} \times 100 + \frac{0.882}{2.5} \times 100 + \frac{0.245}{2.5} \times 100 \right) \\ &= 100 - 48.68 \\ &= 51.32 \% \end{aligned}$$

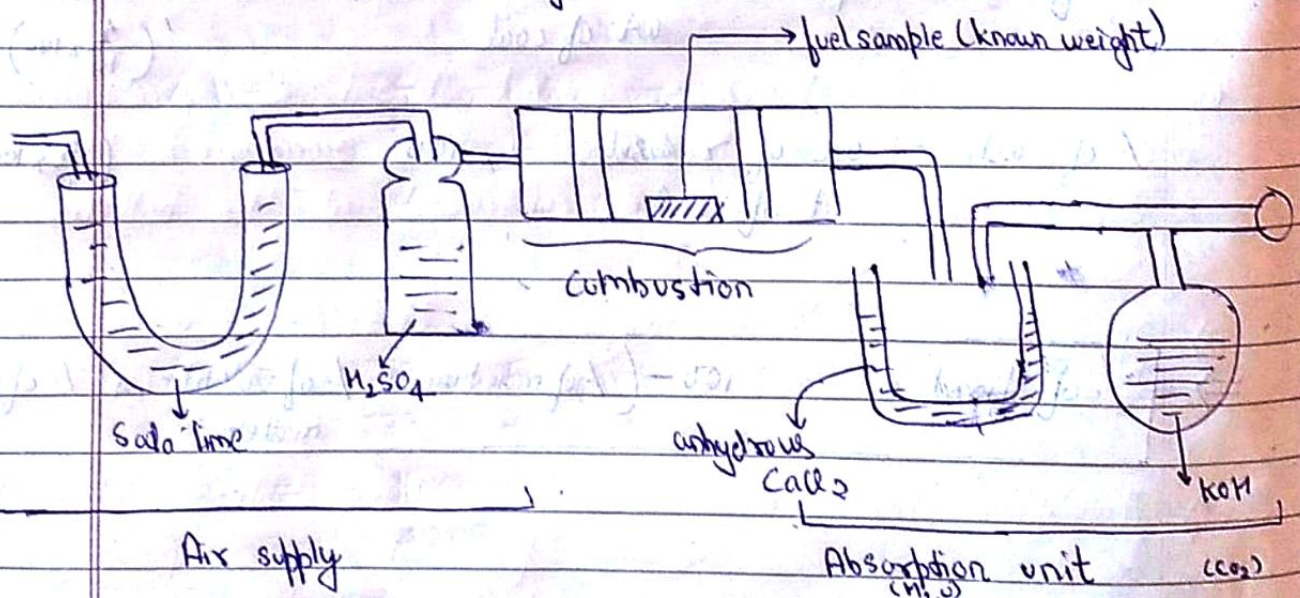
$$\begin{aligned} \% \text{ fixed C (B)} &= 100 - \left(\frac{0.2}{5} \times 100 + \frac{1.42}{5} \times 100 + \frac{0.075}{5} \times 100 \right) \\ &= 100 - 33.9 \\ &= 66.1 \% \end{aligned}$$

(Fuel B is better)

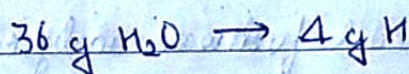
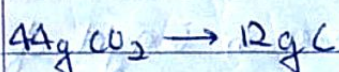
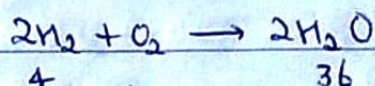
★ Ultimate analysis

% C, % H, % N, % S

Kjeldahl's



Weight of $\text{CO}_2 = x \text{ gms}$
wt. of $\text{H}_2\text{O} = y \text{ gms}$
wt. of fuel = $z \text{ gms}$



$$\Rightarrow \% \text{ of C} = \frac{12 \times x}{44 \times z} \times 100$$

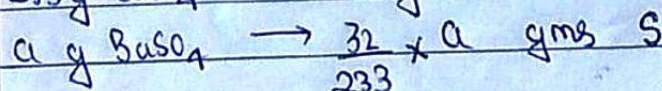
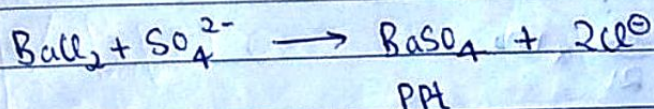
$$\Rightarrow \% \text{ of H} = \frac{4 \times y}{36 \times z} \times 100$$

$\Rightarrow \% \text{ of S}$

Ash $\longrightarrow$ washed $\text{H}_2\text{SO}_4 + \text{BaCl}_2 \longrightarrow \text{BaSO}_4 \xrightarrow{\text{ppt}}$ washed $\downarrow$ dried



weight of BaSO_4



$$\Rightarrow \% S \text{ in fuel} = \frac{32}{233} \times \frac{a}{z} \times 100$$

Q) When coal fuel was analysed the increase in weight of potash bulb was 1.2 gm & CaCl_2 absorb 0.8 gm. This fuel sample was further analysed for sulphur & wt. of BaSO_4 was 230 gm. Determine the %C, H & S when wt. of fuel is 21 gm.

$$\% C = \frac{12 \times 1.2}{44 \times 21} \times 100$$

$$= \frac{1200}{770} = 1.55 \%$$

$$\% H = \frac{4}{36} \times \frac{0.8}{21} \times 100$$

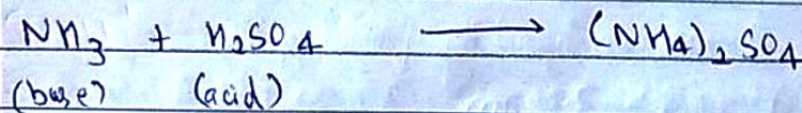
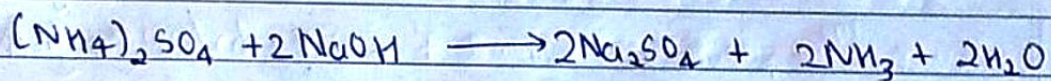
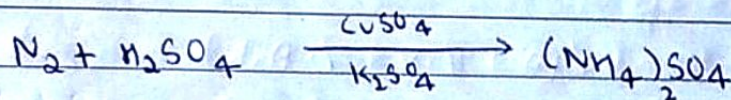
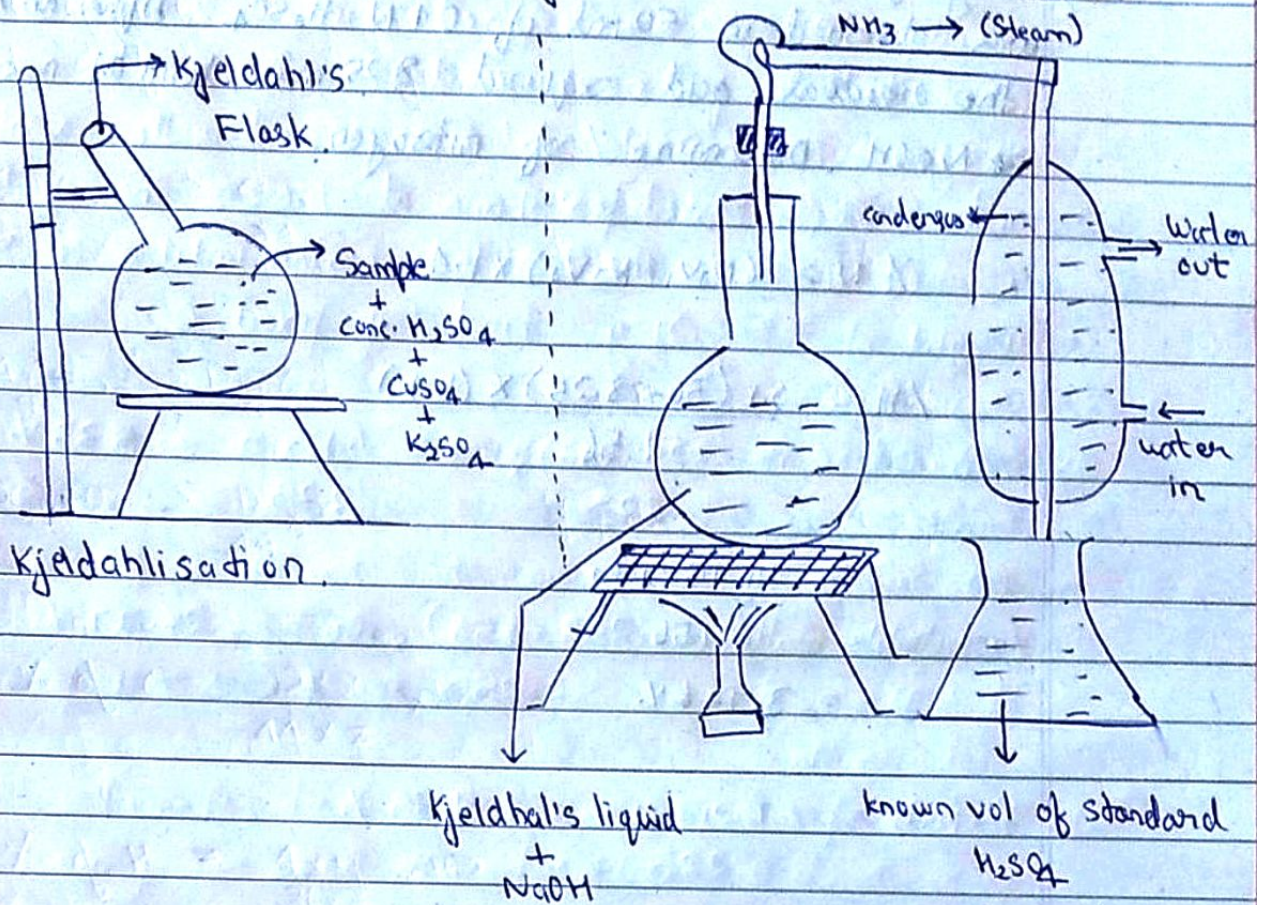
$$= \frac{800}{1890} = 0.42\%$$

$$\% S = \frac{32}{233} \times \frac{50}{21} \times 100$$

$$= \frac{160000}{4893}$$

$$= 32.69\%$$

⇒ Determination of % of Nitrogen (KJELDAHL'S METHOD)



$$\% N = \frac{0.1(V_1 - V_2) \times (1.4)}{W}$$

where normality of H_2SO_4 = normality of $NaOH$ = 0.1 N

W = weight of coal

V_1 = Vol. H_2SO_4 → excess (Total)

V_2 = Vol $NaOH$

$$\%N = \frac{(N_1V_1 - N_2V_2) \times 14 \times 100}{W \times 1000}$$

Q) 1.5g of coal was Jendhalized and ammonia gas evolved was absorbed in 50 ml of 0.1 N H_2SO_4 . After absorption the residual acid required 8.25 ml, point 0.1 normal NaOH. Determine % of nitrogen.

$$\%N = (N_1V_1 - N_2V_2) \times 1.4$$

$$\%N = \frac{0.1(50 - 8.25) \times (1.4)}{1.56}$$

$$= \frac{41.75 \times 1.4}{1.56}$$

$$= 3.746\%$$

Assignment - 1⇒ Proximate analysis

Q1) 2.3 g of air-dried coal sample was taken in a silica crucible. After heating it in an electric oven at $105-110^{\circ}\text{C}$ for 1 hr, the residue weighed ~~0.245 gm~~ 2.2 g. The residue was then ignited at $700-750^{\circ}\text{C}$ to a constant weight of 0.245 gm.

→ In another experiment 1.1 gm of the same coal sample was heated in a silica crucible covered with a vented lid at 925°C for 7 min. After cooling the weight of residue was found to be 0.725 gm. Calculate % of fixed carbon.

$$\text{Wt fuel} = \cancel{2.3 \text{ gm}} 1.1 \text{ gm}$$

$$\% \text{ of moisture} = \frac{(2.3 - 2.2) \times 100}{2.3} = \frac{0.1 \times 100}{2.3} = 4.35 \%$$

$$\% \text{ of ash} = \frac{0.245 \times 100}{2.3} = 10.65 \%$$

$$\text{Initial wt} = 1.1 \text{ gm}$$

$$\text{After heating at } 925^{\circ}\text{C} \text{ wt} = 0.725 \text{ gm}$$

$$\begin{aligned} \% \text{ of volatile matter} &= \frac{W_{\text{lost}} \times 100}{W_{\text{coal}}} = \frac{(1.1 - 0.725) \times 100}{1.1} \\ &= 34.09 - \% \text{ moisture} \\ &= 34.09 - 4.35 \\ &= 29.74 \% \end{aligned}$$

$$\begin{aligned} \% \text{ of fixed C} &= 100 - (\% \text{ m} + \% \text{ v} + \% \text{ Ash}) \\ &= 100 - (4.35 + 29.74 + 10.65) \\ &= 55.26 \% \end{aligned}$$

Q2 A sample of coal was analyzed by following:
 Exactly 2.25 gm was weighed into a silica crucible.
 After heating for one hour at 110°C , the residue weighed 2.13 gm. The crucible next was covered with a vented lid and strongly heated for 7 min at 950°C . The residue weighed 1.55g. The residue heated until a constant weight of 0.215g. Find % of fixed C.

$$\% \text{ moisture} = \frac{(2.25 - 2.13)}{2.25} \times 100 = \frac{0.12 \times 100}{2.25} = 5.33\%$$

$$\% \text{ volatile matter} = \frac{(2.13 - 1.55)}{2.25} \times 100 = 25.77\%$$

$$\% \text{ Ash} = \frac{0.215 \times 100}{2.25} = 9.55\%$$

$$\begin{aligned} \% \text{ of fixed C} &= 100 - (\% \text{ m} + \% \text{ v} + \% \text{ Ash}) \\ &= 100 - (5.33 + 25.77 + 9.55) \\ &= 59.35\% \end{aligned}$$

Q3) A sample of coal was analysed by following:

Exactly Wt. of Coal = 2.00 gm

wt. of loss of moisture = 1.975 g

wt. of loss of V.M = 1.328 g

wt. of Ash = ~~1.328~~ 0.205 g

$$\% \text{ of moisture} = \frac{(2 - 1.975)}{2} \times 100 = 1.25\%$$

$$\% \text{ of vm} = \frac{(1.975 - 1.328)}{2} \times 100 = 32.35\%$$

$$\% \text{ Ash} = \frac{0.205}{2} \times 100 = 10.25\%$$

$$\% \text{ of fixed C} = 100 - [12.5 + 30.35 + 10.25] \\ = 56.15\%$$

⇒ Ultimate analysis & Kjeldahl's method

Q) Calculate C, H, N, S % from the following observation for a sample of coal:

- 2.1 g of coal is burnt in combustion tube. The increase in weight of anhydrous CaCl_2 is 0.53 g and increase in weight of KOH is 5.73 gm.
- 0.75 gm of the coal in Kjeldahl's experiment released NH_3 , which is passed in 50 ml of 0.12 N H_2SO_4 . The H_2SO_4 requires 34 ml of 0.12 N NaOH to neutralize in back titration.
- Washings of Bomb pot when 1.9 gm of coal sample in Bomb calorimeter is treated with BaCl_2 solⁿ to give 0.41 gm BaSO_4 .

$$\% \text{ H} = \frac{\text{Inc. in wt. of } \text{CaCl}_2}{\text{wt. of fuel}} \times \frac{2}{18} \times 100 \\ = \frac{0.53 \times 2 \times 100}{18 \times 2.1} = 2.8\%$$

$$\% \text{ C} = \frac{\text{Inc. in wt. of } \text{KOH} \times 12}{\text{wt. of fuel} \times 44} \times 100 = \frac{5.73 \times 12 \times 100}{44 \times 2.1} \\ = 74.41\%$$

$$\% \text{ S} = \frac{W_{\text{BaSO}_4} \times 32}{233 \times W_{\text{coal}}} \times 100 = \frac{0.41 \times 32 \times 100}{233 \times 1.9} = 2.96\%$$

$$\%N = \frac{\text{Vol. of acid consumed by } NH_3 \times \text{Normality of acid} \times 14}{1000 \times W_{\text{coal}}} \times 100$$

$$\begin{aligned} \%N &= \frac{(50-39) \times 0.12 \times 14 \times 100}{1000 \times 0.75} \\ &= \frac{18.48}{0.75} = 2.46\% \end{aligned}$$

Q) Calculate C, H, S % from the following observation of coal sample:

i) 0.18 gm on burning in combustion chamber in presence of oxygen was found to increase the wt. of $CaCl_2$ by 0.09 gm and that of KOH by 0.5 gm. Calculate the % C & % H present in coal sample.

ii) 0.65 gm of coal was combusted in bomb calorimeter. Washing from bomb on treatment with $BaCl_2$ solⁿ forms 0.031 gm $BaSO_4$ dry ppt.

$$\%H = \frac{0.09 \times 2 \times 100}{18 \times 0.18} = 5.55\%$$

$$\%C = \frac{0.5 \times 12 \times 100}{44 \times 0.18} = 75.75\%$$

$$\%S = \frac{W_{BaSO_4} \times 32}{233 \times W_{\text{coal}}} \times 100$$

$$= \frac{0.031 \times 32}{233} \times \frac{100}{0.65}$$

$$= \frac{992}{1514.5} = 0.65\%$$

Q₁ 0.5 gm of coal sample on complete combustion was found to increase the wt. of CaCl_2 U-tube by 0.2 gm and KOH U-tube by 1.2 gm. Calculate the % C & % H.

$$\begin{aligned}\% \text{H} &= \frac{\text{Inc. in wt. of } \text{CaCl}_2 \times 2 \times 100}{\text{wt. of coal} \times 18} \\ &= \frac{0.2 \times 2 \times 100}{18 \times 0.5} = 4.44\%\end{aligned}$$

$$\% \text{C} = \frac{1.2 \times 12 \times 100}{44 \times 0.5} = 65.45\%$$

⇒ Kjeldahl's method

Q₂ 16 gm of coal sample in Kjeldahl's experiment liberated NH_3 which was absorbed in 50 ml 0.1N H_2SO_4 . The resultant solⁿ required 28 ml of 0.1N NaOH for complete neutralization of H_2SO_4 . Find % N.

$$\begin{aligned}\% \text{N} &= \frac{0.1(V_1 - V_2) \times 14 \times 100}{1000 \times W_{\text{coal}}} \\ &= \frac{0.1 \times 22 \times 14 \times 100}{1000 \times 16} = 1.925\%\end{aligned}$$

Q₃ 1.56 g of coal sample was kjeldahlized & NH_3 gas evolved was absorbed in 50 ml of 0.1N H_2SO_4 . After absorption of excess acid requires 6.25 ml of 0.1N NaOH to neutralise. Find % N.

Let vol of H_2SO_4 used by NaOH be x :-

$$N_1 V_1 = N_2 V_2$$

$$\text{or } x = 6.25$$

$$x = 6.25$$

$$\Rightarrow \text{Acid consumed by } \text{NH}_3 = 50 - 6.25 = 43.75$$

$$\% \text{ N} = \frac{1.4 \times \text{Acid consumed by } \text{NH}_3}{W_{\text{coal}}}$$

$$= \frac{1.4 \times 43.75}{1.56} = 3.926\%$$

Q₂ 1g of sample of coal was used for nitrogen estimation by Kjeldahl's method. The evolved ammonia was converted in 25ml (N/10) H_2SO_4 . To neutralize excess acid, 15 ml of 0.1N NaOH was required. Determine the % N.

$$\% \text{ N} = \frac{0.1 \times 1.4 (25 - 15)}{1}$$

$$= 10 \times 0.1 \times 1.4$$

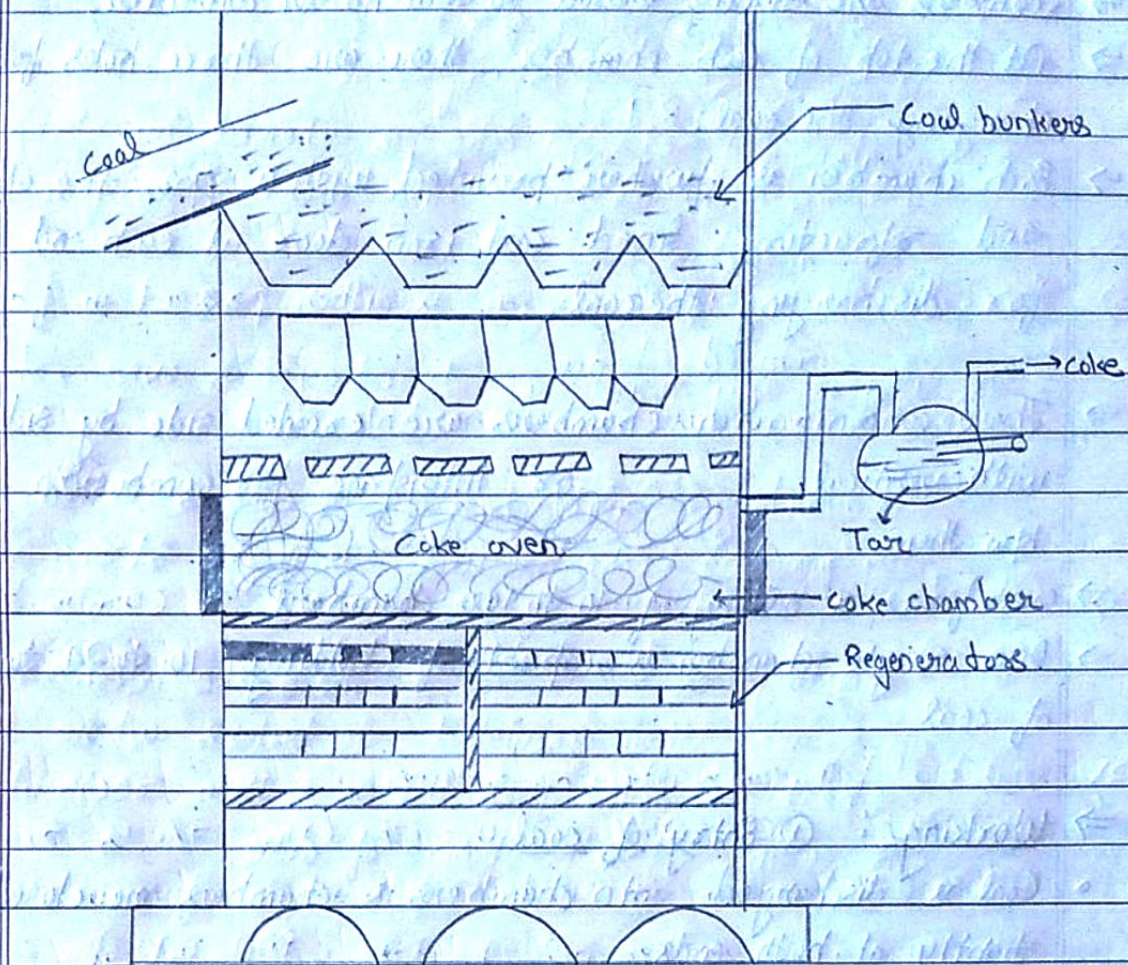
$$= 1.4\%$$

Q₃ 1.5g of coal sample in Kjeldahl's method NH_3 , which was absorbed in 45 ml of 0.1N H_2SO_4 . The resultant soln required 10 ml of 0.1N NaOH for complete neutralisation of H_2SO_4 . Find % of N.

$$\% \text{ of N} = \frac{0.1 (\text{Vol. of } \text{H}_2\text{SO}_4 - \text{Vol. of NaOH}) \times 1.4}{1.5}$$

$$= \frac{0.1 (45 - 10) \times 1.4}{1.5} = 3.266\%$$

★ Otto Hoffmann's Over or By Product over



Single chamber of Otto Hoffmann's over

⇒ Principle : These ovens operate on regeneration principle of heat economy, i.e. utilization of the waste fuel gases for heating of the work of bricks. Moreover, all the valuable products are recovered from the outgoing gases.

⇒ Construction :

Consists of a number of narrow rectangular chambers made up of silica bricks.

→ About the chambers

Usual dimensions : Length - 12 to 14 m Breadth - 5 m
height - 4 to 5 m.

- chambers are tightly closed so that no air can enter.
- At the top of each chamber, there are three holes for charging the coal.
- Each chamber is further provided with a gas take off and refractory lined cast iron door at each end ~~and~~ for discharging the coal.
- These carbonization chambers are created side by side with vertical or interspace for combustion blow them.
- oven forms a battery of 10 to 100 chambers
- One single chamber is capable of holding 10 to 24 tonnes of coal

⇒ Working : ① Entry of coal

- Coal is discharged into chambers & chambers are closed tightly at both ends.
- Ovens are heated to 1200°C by burning gaseous fuel.

② Process of carbonization

- The process takes place layer by layer in the coal charge starting from the two side walls of the oven & moving towards the centre.
- As the coal adjacent to the wall gets heated a plastic zone is form.
- Plastic zone moves from walls towards the central zone & thus, it heats up the central zone.

③ Coal to coke

- As the coal is converted to coke its vol. gets decreased.

- This is because of removal of volatile matter in the form of tar & gas at 500°C .
- At $750 - 800^{\circ}\text{C}$, the plastic mass solidifies into hard and porous mass called coke.

④ Regenerative principle

- To achieve economical heating.
- Waste gases produced during combustion (fuel gases)



passed through regenerators
(Regenerators are built underneath the ovens)



fuel gases pass their heat to the checker brick work of regenerators until the temp rises to 1000°C .



Inlet gases used for combustion are passed through the heated checker brickwork of regenerators.



Inlet gases are preheated before entering into the interspace for combustion.



This process is repeated again & again.

- Thus regenerators work on the principle of alternate heating and cooling cycles.
- This is achieved by periodically changing the direction of flow of gases through the vertical flues every 30 minutes.

⑤ Quenching

- Carbonization of coal takes about 11 to 13 hours.
- After that, red hot coke is pushed outside by means of

a room which is electrically driven.

The coke falls into a quenching car.

- Yield of coke : 75% of coal
- Quenching of red hot coke is achieved by spraying water (wet quenching) or by passing inert gases from boilers (dry quenching)

Q) Why is dry quenching advantageous over wet quenching?

Wet quenching: Sensible heat of the coke is wasted.

Dry quenching: Coke produced is more strong and dense.

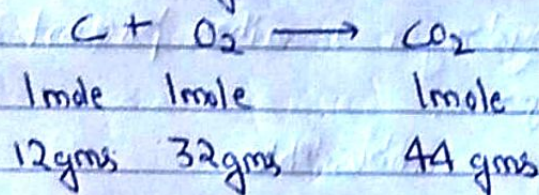
- Inert gases after quenching the coke become heated up & are then circulated to boilers where they pre-heat the boiled fuel water & then recirculated for cooling fresh charge of coke.

★ Combustion

⇒ Calculations of air quantities

Rules :

- 1) Substances always combines in definite proportions.



- 2) 22.4 l of any gas at STP ~~is~~ has always a mass equal to its 1 mole.

$$\begin{aligned}
 22.4 \text{ l O}_2 \text{ at STP} &= 1 \text{ mole} \\
 &= 32 \text{ gms}
 \end{aligned}$$

$$22.4 \text{ l N}_2 \text{ at STP} = 1 \text{ mole} = 28 \text{ gms}$$

- 3) Air contains 21% of Oxygen by volume.
Air contains 23% of oxygen by mass.

$$\Rightarrow 1 \text{ kg of O}_2 \text{ is supplied by} = \frac{100}{23} = 4.35 \text{ kg air}$$

$$\Rightarrow 1 \text{ m}^3 \text{ of O}_2 \text{ is supplied by} = \frac{100}{21} = 4.76 \text{ m}^3 \text{ air}$$

- 4) Molar mass of air = 28.94 g mol⁻¹

- 5) minimum oxygen required for combustion = (Theoretical / Total O₂ requires - O₂ already present in fuel)

- 6) minimum O₂ required for combustion is calculated on the basis of complete combustion.

7) Mass of any gas is converted to its volume at a certain temp & pressure assuming ideal behaviour:

$$\boxed{PV = nRT}$$

atm l K

8) The total amount of hydrogen is either present in free form or combined form.

Combine hydrogen is non-combustible.

10) Calculate weight & volume of air required for combustion of 1 kg of C.

Q2) Gas has foll. composition by volume
 $H_2 = 30\%$, $CO = 20\%$, $CO_2 = 6\%$, $CH_4 = 5\%$, $O_2 = 5\%$, $N_2 = 34\%$

g) 50% excess air is used. Find weight of air actually supplied per m^3 of gas. Mol. wt of air = 28.97 g mol^{-1} .

Ans1)

Constituent	Amount	Combustion eq ⁿ	wt. of O_2 required
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C 1 kg $C + O_2 \rightarrow CO_2$ 2.6 kg

12 gm $\rightarrow$ 32 gm

1 gm $\rightarrow \frac{32}{12}$ gm

$$\left(\frac{32}{12} \times 1000 \right) = \frac{8000}{3} \text{ kg}$$

1 kg $O_2 \rightarrow 4.35 \text{ kg air}$

$\frac{8}{3} \text{ kg } O_2 \rightarrow 2.6 \times 4.35 \text{ kg air}$

= 11.31 kg of air

$$32 \text{ gm } O_2 = 22.4 \text{ l at STP}$$

$$2600 \text{ gm } O_2 = \frac{22.4}{32} \times 2600$$

$$= 1820 \text{ l of } O_2$$

$$1 \text{ l } O_2 = \frac{100}{21} \text{ l of air of vol}^m$$

$$1820 \text{ l of } O_2 = \frac{100}{21} \times 1820 \text{ l air} = 8666 \text{ l}$$

$$= 8.6 \text{ m}^3$$